

Assignment 3

Q1 Create 1 Public Docker Hub registry named flask_webapp_yourname.

- Clone below repository on your system.

<https://github.com/mmumshad/simple-webapp-docker.git>

- Initialize a local repository & copy the code from above repo to your local repository in your working branch.
- Once code is copied to the local repository, build the docker image & add meaningful tags with version 1 and push to Docker Hub registry.
- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , labels app: flask-webapp , containerPort: 8080 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select flask-webapp with port 8080 & targetPort 8080 with any nodePort between range 30000-32768.
- Once a service is created try accessing the web page in the browser as below. (30011 is nodeport mentioned in service.yaml). Meanwhile open app.py from your code to understand paths & output.

http://master_ip:30011/

http://master_ip:30011/how are you

- Now , update the app.py from your code and add below route above if **name == "main"** line

```
@app.route('/Who are you')
```

```
def cloudeithix():
```

```
return 'Yes, I am cloudeithix, and You !!!'
```

- Once the file is updated , rebuild the docker image & add meaningful tags with version 2 and push to Docker Hub registry.
- Now we have the latest docker image in repo, It's time to roll out a new image. Roll out the new Image with all three ways one by one.

1. With kubectl set command

2. With kubectl edit deployment

3. With deployment.yaml file modification.

- Run the # kubectl rollout command to check status and history.
- Note:- Once above step 1 is done , run # kubectl rollout undo deployment command to rollback the change and then try a second way of rollout.
- In the browser run all three routes & notice the changes.

http://master_ip:30011/

http://master_ip:30011/how are you

http://master_ip:30011/Who are you

- Once done with all above steps , commit all the changes to the remote repository

System Output

```
avej_lanjekar@AvejLanjekar:~$ git clone https://github.com/mmumshad/simple-webapp-docker.git
```

```
Cloning into 'simple-webapp-docker'...
```

```
remote: Enumerating objects: 14, done.
```

```
remote: Counting objects: 100% (7/7), done.
```

```
remote: Compressing objects: 100% (5/5), done.
```

```
remote: Total 14 (delta 3), reused 2 (delta 2), pack-reused 7 (from 1)
```

```
Receiving objects: 100% (14/14), done.
```

```
Resolving deltas: 100% (3/3), done.
```

```
avej_lanjekar@AvejLanjekar:~$ cp -r simple-webapp-docker/* flask_webapp/
```

```
avej_lanjekar@AvejLanjekar:~$ ls
```

```
flask_webapp  simple-webapp-docker
```

```
avej_lanjekar@AvejLanjekar:~$ cd flask_webapp/
```

```
avej_lanjekar@AvejLanjekar:~$ ls
```

```
Dockerfile  app.py
```

```
avej_lanjekar@AvejLanjekar:~$ git init
```

```
hint: Using 'master' as the name for the initial branch. This default branch name
```

```
hint: will change to "main" in Git 3.0. To configure the initial branch name
```

```
hint: to use in all of your new repositories, which will suppress this warning,
```

```
hint: call:
```

```
hint:
```

```
hint:   git config --global init.defaultBranch <name>
```

```
hint:
```

```
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
```

```
hint: 'development'. The just-created branch can be renamed via this command:
```

```
hint:
```

```
hint:   git branch -m <name>
```

```
hint:
```

```
hint: Disable this message with "git config set advice.defaultBranchName false"
```

```
Initialized empty Git repository in /home/avej_lanjekar/kubernetes-
```

Assignment/Repetition-1/Assignment3/Q1/flask_webapp/.git/

avej_lanjekar@AvejLanjekar:~ git status

On branch master

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

Dockerfile

app.py

nothing added to commit but untracked files present (use "git add" to track)

avej_lanjekar@AvejLanjekar:~ docker build -t flask_webapp:version1 .

STEP 1/5: FROM ubuntu:20.04

Resolved "ubuntu" as an alias

(/etc/containers/registries.conf.d/shortnames.conf)

Trying to pull docker.io/library/ubuntu:20.04...

Getting image source signatures

Copying blob 13b7e930469f done |

Copying config b7bab04fd9 done |

Writing manifest to image destination

STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip

Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]

Get:2 http://archive.ubuntu.com/ubuntu focal InRelease [265 kB]

Get:3 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64

Packages [32.5 kB]

Get:4 http://security.ubuntu.com/ubuntu focal-security/restricted amd64

Packages [4710 kB]

Get:5 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]

Get:6 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]

Get:7 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages

[1300 kB]

Get:8 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages

[4368 kB]

Get:9 http://archive.ubuntu.com/ubuntu focal/multiverse amd64 Packages [177

kB]

Get:10 http://archive.ubuntu.com/ubuntu focal/universe amd64 Packages [11.3

MB]

avej_lanjekar@AvejLanjekar:~ docker tag flask_webapp:version1

avejlanjekar45/flask_webapp_avej:version1

avej_lanjekar@AvejLanjekar:~ docker push

```
avejlanjekar45/flask_webapp_avej:version1
Getting image source signatures
Copying blob 68199f7ab160 done    |
Copying blob 96a1aa32f272 done    |
Copying blob 7ef1324fc058 done    |
Copying blob 13b7e930469f skipped: already exists
Copying config 0d76e83579 done    |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:~ mkdir kube
```

```
avej_lanjekar@AvejLanjekar:~ cd kube/
```

```
avej_lanjekar@AvejLanjekar:~ vim deployment.yml
```

```
avej_lanjekar@AvejLanjekar:~ ls
```

```
deployment.yml
```

```
avej_lanjekar@AvejLanjekar:~ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: flask-webapp
```

```
  namespace: ns-1
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: flask-webapp
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: flask-webapp
```

```
    spec:
```

```
      containers:
```

```
        - name: flask-webapp
```

```
          image: avejlanjekar45/flask_webapp_avej:version-2
```

```
          ports:
```

```
            - containerPort: 8080
```

```
avej_lanjekar@AvejLanjekar:~ vim service.yml
```

```
avej_lanjekar@AvejLanjekar:~ cat service.yml
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: flask-webapp-svc
```

```
  namespace: ns-1
```

```
spec:
```

```
type: NodePort
selector:
  app: flask-webapp
ports:
  - port: 8080
    targetPort: 8080
    nodePort: 30011
```

avej_lanjekar@AvejLanjekar:~ kubectl get deployments

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
flask-webapp	3/3	3	3	3h15m
nfs-server	1/1	1	1	8h

avej_lanjekar@AvejLanjekar:~ kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
cloudethix-clusterip	ClusterIP	10.97.252.69	<none>	80/TCP
flask-webapp-svc	NodePort	10.104.135.92	<none>	8080:30011/TCP
inventory-service	ClusterIP	10.102.75.180	<none>	5678/TCP
nfs-server	ClusterIP	10.109.166.116	<none>	2049/TCP

avej_lanjekar@AvejLanjekar:~ curl http://192.168.49.2:30011/
Welcome!

avej_lanjekar@AvejLanjekar:~ curl "http://192.168.49.2:30011/how%20are%20you"
I am good, how about you?

avej_lanjekar@AvejLanjekar:~ cat app.py

```
import os
from flask import Flask
app = Flask(__name__)
```

```
@app.route("/")
def main():
    return "Welcome!"
```

```
@app.route('/how are you')
def hello():
    return 'I am good, how about you?'
```

```
if __name__ == "__main__":
    app.run()
```

```
@app.route('/Who are you')
def cloudeithix():
    return 'Yes, I am cloudeithix, and You !!!'
```

```
avej_lanjekar@AvejLanjekar:~ docker build -t flask_webapp:version-2 .
STEP 1/5: FROM ubuntu:20.04
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
--> Using cache
a2753bbdc460a9c05a7240ec9f1be9eaf3732ba6d592b3a403b3ee2b81a82cbf
--> a2753bbdc460
STEP 3/5: RUN pip install flask
--> Using cache
78efa4ef97b780ce79b296f086decd52be0646fd98d8aaf4667f6322cedc4436
--> 78efa4ef97b7
STEP 4/5: COPY app.py /opt/
--> acf77daebf8f
STEP 5/5: ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0 --
port=8080
COMMIT flask_webapp:version-2
--> f62c9c6e657b
Successfully tagged localhost/flask_webapp:version-2
f62c9c6e657bbac895b6de5dd6a831b41298094116dc27c5bb69ad3977a3910f
```

```
avej_lanjekar@AvejLanjekar:~ docker tag flask_webapp:version-2
avejlanjekar45/flask_webapp_avej:version-2
```

```
avej_lanjekar@AvejLanjekar:~ docker push
avejlanjekar45/flask_webapp_avej:version-2
Getting image source signatures
Copying blob 0cb734515e39 done    |
Copying blob 67f28aede3f8 skipped: already exists
Copying blob 13b7e930469f skipped: already exists
Copying blob c6459ed4af3c skipped: already exists
Copying config f62c9c6e65 done    |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:~ kubectl set image deployment/flask-webapp flask-
webapp=avejlanjekar45/flask_webapp_avej:version-2
deployment.apps/flask-webapp image updated
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
avej_lanjekar@AvejLanjekar:~ curl http://192.168.49.2:30011/
Welcome!
```

```
avej_lanjekar@AvejLanjekar:~ curl http://192.168.49.2:30011/how%20are%20you
I am good, how about you?
```

```
avej_lanjekar@AvejLanjekar:~ curl "http://192.168.49.2:30011/Who%20are%20you"
Yes, I am cloudethix, and You !!!
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
```

```
avej_lanjekar@AvejLanjekar:~ kubectl edit deployment/flask-webapp
deployment.apps/flask-webapp edited
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
2          <none>
3          <none>
```

```
avej_lanjekar@AvejLanjekar:~ kubectl apply -f deployment.yml
deployment.apps/flask-webapp configured
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
avej_lanjekar@AvejLanjekar:~ kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
3          <none>
4          <none>
```

```
avej_lanjekar@AvejLanjekar:~ curl "http://192.168.49.2:30011/how%20are%20you"
I am good, how about you?
```

```
avej_lanjekar@AvejLanjekar:~ curl "http://192.168.49.2:30011/Who%20are%20you"
Yes, I am cloudethix, and You !!!
```

```
avej_lanjekar@AvejLanjekar:~ curl "http://192.168.49.2:30011/"
Welcome!
```

```

avej_lanjekar@AvejLanjekar:~ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    app.py
    kube/

nothing added to commit but untracked files present (use "git add" to track)

avej_lanjekar@AvejLanjekar:~ git add .

avej_lanjekar@AvejLanjekar:~ git commit -m "Added kubernetes deployment"
[master (root-commit) e6006b7] Added kubernetes deployment
 4 files changed, 60 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 kube/deployment.yml
 create mode 100644 kube/service.yml

avej_lanjekar@AvejLanjekar:~ git status
On branch master
nothing to commit, working tree clean

avej_lanjekar@AvejLanjekar:~ git remote add origin
git@github.com:Avejlanjekar/flask-k8s-rolling-update-deployment.git

avej_lanjekar@AvejLanjekar:~ git branch -M main

avej_lanjekar@AvejLanjekar:~ git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.05 KiB | 269.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Avejlanjekar/flask-k8s-rolling-update-deployment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```


- Download and install Lens & access your k8s cluster from Lens.
- Create nginx Pod and Nodeport service. Check the Pod logs from Lens.
- Check the service from lens. Also login to the pod shell using the lens.
- Take snaps and delete the resources that you have just created.

```
avej_lanjekar@AvejLanjekar:Q4$ minikube dashboard
🔧 Enabling dashboard ...
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all
features please run:
```

```
minikube addons enable metrics-server
```

```
😬 Verifying dashboard health ...
🚀 Launching proxy ...
😬 Verifying proxy health ...
🎉 Opening http://127.0.0.1:43941/api/v1/namespaces/kubernetes-
dashboard/services/http:kubernetes-dashboard:/proxy/ in your default
browser...
👉 http://127.0.0.1:43941/api/v1/namespaces/kubernetes-
dashboard/services/http:kubernetes-dashboard:/proxy/
```

```
avej_lanjekar@AvejLanjekar:Q2$ ls
pod.yml  service.yml
```

```
avej_lanjekar@AvejLanjekar:Q2$ cat pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: pod-nginx
  labels:
    app: nginx
spec:
  containers:
    - name: nginx
      image: nginx
      ports:
        - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:Q2$ cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: nodeport-svc
```

```
spec:
  selector:
    app: nginx
  type: NodePort
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

Q3 Create 1 Public Docker Hub registry named **cloudethix_configmap_yourname**.

- Clone below repository on your system.
<https://github.com/zembutsu/docker-sample-nginx.git>
- Initialize a local repository & copy the code from above repo to your local repository in the working branch.
- Once code is copied , build a docker image from docker file and add meaningful tags and push to docker hub repository.
- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , labels app: frontend-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select frontend-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Once the service is created try accessing the web page in the browser as below. Notice the changes & take the snap.
- Now create a configmap.yaml file with below data & delete the deployment that you have created.

```
<html>
<body>
<h1> I am Cloudethix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
```

- Then update the same deployment.yaml file and mount configmap as volume on container using volumeMounts with mountPath /usr/share/nginx/html/

- Now it's time to create configmap & deployment. Once created , try to access the webpage in the browser & confirm that the index page is the same as we have in configmap.

```
avej_lanjekar@AvejLanjekar:~ git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
```

```
avej_lanjekar@AvejLanjekar:~ mkdir clouethix-configmap
```

```
avej_lanjekar@AvejLanjekar:~ ls
clouethix-configmap  docker-sample-nginx
```

```
avej_lanjekar@AvejLanjekar:~ cp docker-sample-nginx/* clouethix-configmap/
```

```
avej_lanjekar@AvejLanjekar:~ ls
clouethix-configmap  docker-sample-nginx
```

```
avej_lanjekar@AvejLanjekar:~ cd clouethix-configmap/
```

```
avej_lanjekar@AvejLanjekar:~ ls
Dockerfile  LICENSE  README.md  default.conf  index.html
```

```
avej_lanjekar@AvejLanjekar:~ docker build -t clouethix-configmap .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
0989e7142eba649abc4d95f1780a91bf1f19479e0fc6153f85667cfc7b500de5
--> 0989e7142eba
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT clouethix-configmap
--> b8bf8728f7f5
Successfully tagged localhost/clouethix-configmap:latest
b8bf8728f7f56726509964c82b8020ce24f4c0cd049539aeb69c604660d03136
```

```
avej_lanjekar@AvejLanjekar:~ docker tag clouethix-configmap
avejlanjekar45/clouethix_configmap_avej
```

```
avej_lanjekar@AvejLanjekar:~ docker push
avejlanjekar45/clouethix_configmap_avej
```

```
Getting image source signatures
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 8cdf4f23778 skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob 0d62d88506ba skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob 03c47411bc64 done    |
Copying blob da4dea1b00af skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying blob ba05887de274 skipped: already exists
Copying config b8bf8728f7 done    |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:~ cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudethix-configmap
  namespace: ns-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp
  template:
    metadata:
      labels:
        app: frontend-webapp
    spec:
      containers:
        - name: cloudethix-cm
          image: avejlanjekar45/cloudethix_configmap_avej
          ports:
            - containerPort: 80
          volumeMounts:
            - name: frontend-config-map
              mountPath: /usr/share/nginx/html/
      volumes:
        - name: frontend-config-map
          configMap:
            name: frontend-config
```

```
avej_lanjekar@AvejLanjekar:~ cat service.yaml
apiVersion: v1
```

```
kind: Service
metadata:
  name: cloudethix-configmap-svc
  namespace: ns-1
spec:
  type: NodePort
  selector:
    app: frontend-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010
```

```
avej_lanjekar@AvejLanjekar:~ curl http://192.168.49.2:30010
<html>
<body>
  <h1>Host: cloudethix-configmap-7b8849c65f-d4464</h1>
  Version: 1.1
</body>
</html>
```

```
avej_lanjekar@AvejLanjekar:~ kubectl delete deployment cloudethix-configmap
deployment.apps "cloudethix-configmap" deleted from ns-1 namespace
```

```
avej_lanjekar@AvejLanjekar:~ cat configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: frontend-config
data:
  index.html: |
    <html>
    <body>
    <h1>I am Cloudethix Team, Are you ?!!</h1>
    Version: 1.1
    </body>
    </html>
```

```
avej_lanjekar@AvejLanjekar:~ kubectl apply -f configmap.yaml
configmap/frontend-config created
```

```
avej_lanjekar@AvejLanjekar:~ cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudethix-configmap
```

```

namespace: ns-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp
  template:
    metadata:
      labels:
        app: frontend-webapp
    spec:
      containers:
        - name: cloudethix-cm
          image: avejlanjekar45/cloudethix_configmap_avej
          ports:
            - containerPort: 80
          volumeMounts:
            - name: frontend-config-map
              mountPath: /usr/share/nginx/html/
      volumes:
        - name: frontend-config-map
          configMap:
            name: frontend-config

```

```
avej_lanjekar@AvejLanjekar:~ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
cloudethix-configmap	3/3	3	3	5m18s

```
avej_lanjekar@AvejLanjekar:~ kubectl get cm
```

NAME	DATA	AGE
frontend-config	1	5m53s

```
avej_lanjekar@AvejLanjekar:~ curl http://192.168.49.2:30010
```

```

<html>
<body>
<h1>I am Cloudethix Team, Are you ?!!</h1>
Version: 1.1
</body>
</html>

```

Que 4 →

- Create 1 Public Docker Hub registry named cloudethix_multicontainer_yourname.
- Clone below repository on your system.

<https://github.com/janakiramm/Kubernetes-multi-container-pod.git>

- Initialize a local repository & copy the code from above repo to your local repository in any of your working branches.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Now go to the deploy directory and notice the files.
- Here, web-pod-1.yml file will create the pod with two containers (Multi container). Take a note of labels , name of containers and ports. Also, please make sure you will update the python container image that you have pushed to your docker registry.
- Now, open web-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Now open db-pod.yml & notice the labels , name , Image, containerPort and apply the file.
- Now open the db-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Once above files are applied , Check that the Pods and Services are created using command line or lens.

- Now , from the command line run below urls & notice the changes.

`curl http://NODE_IP :NODE_PORT/init`

Initialize the database with sample schema

- Now it's time to Insert some sample data. Make sure you will use correct

`NODE_IP :NODE_PORT`

`curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",`

`"user": "John Doe"}' http://NODE_IP :NODE_PORT/users/add`

`curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",`

`"user": "Jane Doe"}' http://NODE_IP :NODE_PORT/users/add`

`curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",`

`"user": "Bill Colls"}' http://NODE_IP :NODE_PORT/users/add`

`curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",`

`"user": "Mike Taylor"}' http://NODE_IP :NODE_PORT/users/add`

- Now access the data that we have added to database using below command.

`curl http://NODE_IP :NODE_PORT/users/1`

- The second time you access the data, it appends '(c)' indicating that it is pulled from the Redis cache.

- Also, try to access mysql shell i.e db pod & run `select * from the users table`. check app.py for DB related information.

- Prepare proper documentation in brief & write start to end flow. Refer below link if you face any issues.

<https://github.com/janakiramm/Kubernetes-multi-container-pod>

```
avej_lanjekar@AvejLanjekar: git clone
https://github.com/janakiramm/Kubernetes-multi-container-pod.git
Cloning into 'Kubernetes-multi-container-pod'...
remote: Enumerating objects: 51, done.
remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)
Receiving objects: 100% (51/51), 88.14 KiB | 5.88 MiB/s, done.
Resolving deltas: 100% (21/21), done.

avej_lanjekar@AvejLanjekar:Q4$ ls
Kubernetes-multi-container-pod

avej_lanjekar@AvejLanjekar:Q4$ mkdir multi-container
avej_lanjekar@AvejLanjekar:Q4$ cd multi-container/

avej_lanjekar@AvejLanjekar:Q4$ cp -r Kubernetes-multi-container-pod/* multi-
container/

avej_lanjekar@AvejLanjekar:Q4$ cd multi-container/
avej_lanjekar@AvejLanjekar:multi-container$ ls
Build  Deploy  README.md  multi-container-pod.png

avej_lanjekar@AvejLanjekar:multi-container$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /mnt/c/Project-Nexus/KUBE/kubernetes-
Assignment/Assignment3/Q4/multi-container/.git/

avej_lanjekar@AvejLanjekar:multi-container$ git status
On branch master

No commits yet
```


Untracked files:

(use "git add <file>..." to include in what will be committed)

Build/

Deploy/

README.md

multi-container-pod.png

nothing added to commit but untracked files present (use "git add" to track)

avej_lanjekar@AvejLanjekar: cd Build/

avej_lanjekar@AvejLanjekar:Build\$ ls

Dockerfile app.py docker-compose.yml requirements.txt

avej_lanjekar@AvejLanjekar:Build\$ docker build -t multi-container .

WARN[0000] "/" is not a shared mount, this could cause issues or missing mounts with rootless containers

STEP 1/3: FROM python:2.7-onbuild

STEP 2/6: COPY requirements.txt /usr/src/app/

--> 0bc887bdeb7e

STEP 3/6: RUN pip install --no-cache-dir -r requirements.txt

Collecting flask (from -r requirements.txt (line 1))

Downloading

<https://files.pythonhosted.org/packages/e8/6d/994208daa354f68fd89a34a8bafbeaab26fda84e7af1e35bdaed02b667e6/Flask-1.1.4-py2.py3-none-any.whl> (94kB)

Collecting redis (from -r requirements.txt (line 2))

Downloading

<https://files.pythonhosted.org/packages/a7/7c/24fb0511df653cf1a5d938d8f5d19802a88cef255706fdda242ff97e91b7/redis-3.5.3-py2.py3-none-any.whl> (72kB)

Collecting mysql-python (from -r requirements.txt (line 3))

Downloading

<https://files.pythonhosted.org/packages/a5/e9/51b544da85a36a68debe7a7091f068d802fc515a3a202652828c73453cad/MySQL-python-1.2.5.zip> (108kB)

Collecting itsdangerous<2.0,>=0.24 (from flask->-r requirements.txt (line 1))

Downloading

<https://files.pythonhosted.org/packages/76/ae/44b03b253d6fade317f32c24d100b3b35c2239807046a4c953c7b89fa49e/itsdangerous-1.1.0-py2.py3-none-any.whl>

Collecting click<8.0,>=5.1 (from flask->-r requirements.txt (line 1))

Downloading

<https://files.pythonhosted.org/packages/d2/3d/fa76db83bf75c4f8d338c2fd15c8d33fdd7ad23a9b5e57eb6c5de26b430e/click-7.1.2-py2.py3-none-any.whl> (82kB)

Collecting Werkzeug<2.0,>=0.15 (from flask->-r requirements.txt (line 1))

Downloading

<https://files.pythonhosted.org/packages/cc/94/5f7079a0e00bd6863ef8f1da638721e9da21e5bacee597595b318f71d62e/Werkzeug-1.0.1-py2.py3-none-any.whl> (298kB)

Collecting Jinja2<3.0,>=2.10.1 (from flask->-r requirements.txt (line 1))

Downloading

```
https://files.pythonhosted.org/packages/7e/c2/1eece8c95ddbc9b1aeb64f5783a9e07a
286de42191b7204d67b7496ddf35/Jinja2-2.11.3-py2.py3-none-any.whl (125kB)
Collecting MarkupSafe>=0.23 (from Jinja2<3.0,>=2.10.1->flask->-r
requirements.txt (line 1))
  Downloading
https://files.pythonhosted.org/packages/fb/40/f3adb7cf24a8012813c5edb20329eb22
d5d8e2a0ecf73d21d6b85865da11/MarkupSafe-1.1.1-cp27-cp27mu-
manylinux1_x86_64.whl
Installing collected packages: itsdangerous, click, Werkzeug, MarkupSafe,
Jinja2, flask, redis, mysql-python
  Running setup.py install for mysql-python: started
    Running setup.py install for mysql-python: finished with status 'done'
Successfully installed Jinja2-2.11.3 MarkupSafe-1.1.1 Werkzeug-1.0.1 click-
7.1.2 flask-1.1.4 itsdangerous-1.1.0 mysql-python-1.2.5 redis-3.5.3
You are using pip version 10.0.1, however version 20.3.4 is available.
You should consider upgrading via the 'pip install --upgrade pip' command.
--> 6612ac212805
STEP 4/6: COPY . /usr/src/app
--> 7f3c6fd69d9b
STEP 5/6: EXPOSE 5000
--> bd92e248818b
STEP 6/6: CMD [ "python", "app.py" ]
COMMIT multi-container
--> 2d16b8e25afb
Successfully tagged localhost/multi-container:latest
```

```
avej_lanjekar@AvejLanjekar:Build$ docker tag multi-container
avejlanjekar45/cloudethix_multicontainer_avej
```

```
avej_lanjekar@AvejLanjekar:Build$ docker push
avejlanjekar45/cloudethix_multicontainer_avej
Getting image source signatures
Copying blob 6ebaeb074589 skipped: already exists
Copying blob 46dde23c37b3 skipped: already exists
Copying blob 56f10ddf1173 skipped: already exists
Copying blob d660b1f15b9b skipped: already exists
Copying blob 0c3de61682aa skipped: already exists
Copying blob e7428f935583 skipped: already exists
Copying blob 28f1ec820014 done    |
Copying blob 3932b6e7ee59 done    |
Copying blob 487e192646e2 done    |
Copying blob 3de1c6ceef68 skipped: already exists
Copying blob 3106f7df3d1c skipped: already exists
Copying blob 4473537c621d skipped: already exists
Copying config 2d16b8e25a done    |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:Build$ cd ..
```

```
avej_lanjekar@AvejLanjekar: ls
```

```
Build Deploy README.md multi-container-pod.png
```

```
avej_lanjekar@AvejLanjekar: cd Deploy/
```

```
avej_lanjekar@AvejLanjekar:Deploy$ ls
```

```
db-pod.yml db-svc.yml web-pod-1.yml web-pod-2.yml web-rc.yml web-svc.yml
```

```
avej_lanjekar@AvejLanjekar:Deploy$ cat web-pod-1.yml
```

```
apiVersion: "v1"
```

```
kind: Pod
```

```
metadata:
```

```
  name: web1
```

```
  labels:
```

```
    name: web
```

```
    app: demo
```

```
spec:
```

```
  containers:
```

```
    - name: redis
```

```
      image: redis
```

```
      ports:
```

```
        - containerPort: 6379
```

```
          name: redis
```

```
          protocol: TCP
```

```
    - name: python
```

```
      image: avejlanjekar45/cloudethix_multicontainer_avej
```

```
      env:
```

```
        - name: "REDIS_HOST"
```

```
          value: "localhost"
```

```
      ports:
```

```
        - containerPort: 5000
```

```
          name: http
```

```
          protocol: TCP
```

```
avej_lanjekar@AvejLanjekar:Deploy$ cat web-svc.yml
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: web
```

```
  labels:
```

```
    name: web
```

```
    app: demo
```

```
spec:
```

```
selector:
  name: web
type: NodePort
ports:
- port: 80
  name: http
  targetPort: 5000
  protocol: TCP
```

```
avej_lanjekar@AvejLanjekar:Deploy$ kubectl apply -f web-pod-1.yml
pod/web1 created
```

```
avej_lanjekar@AvejLanjekar:Deploy$ kubectl apply -f web-svc.yml
service/web created
```

```
avej_lanjekar@AvejLanjekar:Deploy$ cat db-pod.yml
```

```
apiVersion: "v1"
kind: Pod
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  containers:
  - name: mysql
    image: mysql:5.7.25
    ports:
      - containerPort: 3306
        protocol: TCP
    env:
      -
        name: "MYSQL_ROOT_PASSWORD"
        value: "password"
```

```
avej_lanjekar@AvejLanjekar:Deploy$ kubectl apply -f db-pod.yml
pod/mysql created
```

```
avej_lanjekar@AvejLanjekar:Deploy$ cat db-svc.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
```

```
spec:
  ports:
    - port: 3306
      name: mysql
      targetPort: 3306
  selector:
    name: mysql
    app: demo
```

```
avej_lanjekar@AvejLanjekar:Deploy$ kubectl apply -f db-svc.yml
service/mysql configured
```

```
avej_lanjekar@AvejLanjekar:Deploy$ minikube service web --url
http://127.0.0.1:46311
```

! Because you are using a Docker driver on linux, the terminal needs to be open to run it.

```
avej_lanjekar@AvejLanjekar:~$ curl http://127.0.0.1:46311/init
DB Init done
```

```
avej_lanjekar@AvejLanjekar:~$ curl -i -H "Content-Type: application/json" -X
POST -d '{"uid": "2",
"user":"Jane Doe"}' http://127.0.0.1:46311/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 05:42:48 GMT
```

```
Addedavej_lanjekar@AvejLanjekar:~$
avej_lanjekar@AvejLanjekar:~$ curl -i -H "Content-Type: application/json" -X
POST -d '{"uid": "3",
"user":"Bill Colls"}' http://127.0.0.1:46311/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 05:43:13 GMT
```

```
avej_lanjekar@AvejLanjekar:~$ curl -i -H "Content-Type: application/json" -X
POST -d '{"uid": "4",
"user":"Mike Taylor"}' http://127.0.0.1:46311/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
```

Date: Thu, 03 Sep 2026 05:43:34 GMT

```
avej_lanjekar@AvejLanjekar:~$ curl http://127.0.0.1:46311/users/1
John Doe
```

```
avej_lanjekar@AvejLanjekar:~$ curl http://127.0.0.1:46311/users/1
John Doe(c)
```

```
avej_lanjekar@AvejLanjekar:~$ kubectl exec -it mysql -- mysql -uroot -
ppassword
```

```
mysql: [Warning] Using a password on the command line interface can be
insecure.
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 4
```

```
Server version: 5.7.25 MySQL Community Server (GPL)
```

```
Copyright (c) 2000, 2019, Oracle and/or its affiliates. All rights reserved.
```

```
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> show databases;
```

```
+-----+
```

```
| Database |
```

```
+-----+
```

```
| information_schema |
```

```
| USERDB |
```

```
| mysql |
```

```
| performance_schema |
```

```
| sys |
```

```
+-----+
```

```
5 rows in set (0.01 sec)
```

```
mysql> use information_schema
```

```
Reading table information for completion of table and column names
```

```
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
```

```
mysql>
```

```
mysql> show tables;
```

```
+-----+
```

```
| Tables_in_information_schema |
```

```
+-----+
```

CHARACTER_SETS	
COLLATIONS	
COLLATION_CHARACTER_SET_APPLICABILITY	
COLUMNS	
COLUMN_PRIVILEGES	
ENGINES	
EVENTS	
FILES	
GLOBAL_STATUS	
GLOBAL_VARIABLES	
KEY_COLUMN_USAGE	
OPTIMIZER_TRACE	
PARAMETERS	
PARTITIONS	
PLUGINS	
PROCESSLIST	
PROFILING	
REFERENTIAL_CONSTRAINTS	
ROUTINES	
SCHEMATA	
SCHEMA_PRIVILEGES	
SESSION_STATUS	
SESSION_VARIABLES	
STATISTICS	
TABLES	
TABLESPACES	
TABLE_CONSTRAINTS	
TABLE_PRIVILEGES	
TRIGGERS	
USER_PRIVILEGES	
VIEWS	
INNODB_LOCKS	
INNODB_TRX	
INNODB_SYS_DATAFILES	
INNODB_FT_CONFIG	
INNODB_SYS_VIRTUAL	
INNODB_CMP	
INNODB_FT_BEING_DELETED	
INNODB_CMP_RESET	
INNODB_CMP_PER_INDEX	
INNODB_CMPMEM_RESET	
INNODB_FT_DELETED	
INNODB_BUFFER_PAGE_LRU	
INNODB_LOCK_WAITS	
INNODB_TEMP_TABLE_INFO	
INNODB_SYS_INDEXES	

INNODB_SYS_TABLES
INNODB_SYS_FIELDS
INNODB_CMP_PER_INDEX_RESET
INNODB_BUFFER_PAGE
INNODB_FT_DEFAULT_STOPWORD
INNODB_FT_INDEX_TABLE
INNODB_FT_INDEX_CACHE
INNODB_SYS_TABLESPACES
INNODB_METRICS
INNODB_SYS_FOREIGN_COLS
INNODB_CMPMEM
INNODB_BUFFER_POOL_STATS
INNODB_SYS_COLUMNS
INNODB_SYS_FOREIGN
INNODB_SYS_TABLESTATS

+-----+

61 rows in set (0.00 sec)

mysql> select * from users;

ERROR 1109 (42S02): Unknown table 'users' in information_schema

mysql> use USERDB;

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

mysql>

mysql>

mysql> show tables;

Tables_in_USERDB
users

+-----+

1 row in set (0.00 sec)

mysql> select * from users;

ID	USER
1	John Doe
2	Jane Doe
3	Bill Colls
4	Mike Taylor

+-----+

4 rows in set (0.00 sec)


```
mysql>
```

Que 5 →

- Create 1 Public Docker Hub registry named cloudehix_Initcontainer_yourname.
 - Clone below repository on your system.
<https://github.com/janakiramm/simpleapp.git>
 - Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
 - Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
 - Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , label app: simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
 - Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
 - Open the webpage in the browser and notice the changes and capture the snap.
 - Then delete the deployment that you have just created.
 - Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.
 - Once above changes are added, add initContainers block with below parameters. Also add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {} volume.
- initContainers:

- name: install
image: busybox:1.28
command:
 - wget
 - "-O"
 - "/work-dir/index.html"

- <http://info.cern.ch>
volumeMounts:
- name: workdir
mountPath: "/work-dir"
 - Add volumes with emptyDir: {} in deployment.yaml file.
 - Once the deployment.yaml file is ready, create the deployment & access the page in the browser and notice the changes.

```
avej_lanjekar@AvejLanjekar: git clone
https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
remote: Enumerating objects: 47, done.
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
Receiving objects: 100% (47/47), 8.20 KiB | 2.05 MiB/s, done.
Resolving deltas: 100% (9/9), done.

avej_lanjekar@AvejLanjekar: ls
simpleapp

avej_lanjekar@AvejLanjekar:Q5$ mkdir simple-web-app

avej_lanjekar@AvejLanjekar:Q5$ ls
simple-web-app  simpleapp

avej_lanjekar@AvejLanjekar:Q5$ cp -r simpleapp/* simple-web-app/

avej_lanjekar@AvejLanjekar:Q5$ ls
simple-web-app  simpleapp

avej_lanjekar@AvejLanjekar:Q5$ cd simple-web-app/

avej_lanjekar@AvejLanjekar:simple-web-app$ ls
Dockerfile  html  wrapper.sh

avej_lanjekar@AvejLanjekar:simple-web-app$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
```

```
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/avej_lanjekar/kubernetes-
Assignment/Repetition-1/Assignment3/Q5/simple-web-app/.git/
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    html/
    wrapper.sh

nothing added to commit but untracked files present (use "git add" to track)

avej_lanjekar@AvejLanjekar:simple-web-app$ ls
Dockerfile  html  wrapper.sh

avej_lanjekar@AvejLanjekar:simple-web-app$ docker build -t init-cont .
STEP 1/4: FROM nginx
STEP 2/4: COPY wrapper.sh /
--> Using cache
b6124d5e3dd17e9cf92678765ea94fa2786cc2ff821e1da52bb594a1e9a9be96
--> b6124d5e3dd1
STEP 3/4: COPY html /usr/share/nginx/html
--> Using cache
dcb12e28f080d5b068f4eb98397ac7d1ab400700104cabb97757e65959d2b7f8
--> dcb12e28f080
STEP 4/4: CMD ["/wrapper.sh"]
--> Using cache
3ba2380d007b1284837510a68d05033921f172a6bdc55190674f85be3f5db0cf
COMMIT init-cont
--> 3ba2380d007b
Successfully tagged localhost/init-cont:latest
Successfully tagged
localhost/avejlanjekar45/cloudethix_initcontainer_avej:latest
```

```
Successfully tagged localhost/initcontainer:latest
3ba2380d007b1284837510a68d05033921f172a6bdc55190674f85be3f5db0cf
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$ docker tag init-cont
avejlanjekar45/cloudethix_initcontainer_avej
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$
avej_lanjekar@AvejLanjekar:simple-web-app$ docker push
avejlanjekar45/cloudethix_initcontainer_avej
Getting image source signatures
Copying blob 657dd7fba849 skipped: already exists
Copying blob c90544874aaf skipped: already exists
Copying blob 30576ad53d33 skipped: already exists
Copying blob b8f66660faa6 skipped: already exists
Copying blob 8f655e1bd5c1 skipped: already exists
Copying blob 6310eb16bf42 skipped: already exists
Copying blob f2700ca67725 skipped: already exists
Copying blob ab9312376d35 skipped: already exists
Copying blob 0a35a4e59186 skipped: already exists
Copying config 3ba2380d00 done |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:~$ mkdir kube
```

```
avej_lanjekar@AvejLanjekar:~$ cd kube/
```

```
avej_lanjekar@AvejLanjekar:~$ kube$ cat deployment.yml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: init-cont-deplo
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp
  template:
    metadata:
      labels:
        app: simpleapp-webapp
    spec:
      containers:
        - name: init-cont
          image: avejlanjekar45/cloudethix_initcontainer_avej
```

```
ports:
  - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:kube$ cat service.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: init-service
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 32000
```

```
avej_lanjekar@AvejLanjekar:kube$ kubectl apply -f service.yml
service/init-service created
```

```
avej_lanjekar@AvejLanjekar:kube$ kubectl get svc init-service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
init-service	NodePort	10.99.78.194	<none>	80:32000/TCP	9s

```
avej_lanjekar@AvejLanjekar:kube$ minikube ip
192.168.49.2
```

```
avej_lanjekar@AvejLanjekar:kube$ curl http://192.168.49.2:32000
```

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Sample Deployment</title>
  <style>
    body {
      color: #ffffff;
      background-color: blue;
      font-family: Arial, sans-serif;
      font-size: 14px;
    }

    h1 {
      font-size: 500%;
      font-weight: normal;
      margin-bottom: 0;
```

```

    }

    h2 {
        font-size: 200%;
        font-weight: normal;
        margin-bottom: 0;
    }
</style>
</head>
<body>
    <div align="center">
        <h1>Welcome to V1 of the web application</h1>
        <h2>This application will be deployed on Kubernetes.</h2>
    </div>
</body>
</html>

```

```

avej_lanjekar@AvejLanjekar:kube$ kubectl delete -f deployment.yml
deployment.apps "init-cont-deplo" deleted from default namespace

```

```

avej_lanjekar@AvejLanjekar:kube$ vim deployment.yml

```

```

avej_lanjekar@AvejLanjekar:kube$ cat deployment.yml

```

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: init-cont-deplo
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp

  template:
    metadata:
      labels:
        app: simpleapp-webapp
    spec:
      initContainers:
        - name: install
          image: busybox:1.28
          command:
            - wget
            - -O
            - /work-dir/index.html
            - http://info.cern.ch
          volumeMounts:

```

```

      - name: workdir
        mountPath: /work-dir

containers:
  - name: init-cont
    image: avejlanjekar45/cloudethix_initcontainer_avej
    ports:
      - containerPort: 80
    volumeMounts:
      - name: workdir
        mountPath: /usr/share/nginx/html
volumes:
  - name: workdir
    emptyDir: {}

```

```

avej_lanjekar@AvejLanjekar:kube$ kubectl apply -f deployment.yml
deployment.apps/init-cont-deplo created

```

```

avej_lanjekar@AvejLanjekar:kube$ curl http://192.168.49.2:32000
<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

<h1>http://info.cern.ch - home of the first website</h1>
<p>From here you can:</p>
<ul>
<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>
<li><a href="http://line-
mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
the line-mode browser simulator</a></li>
<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>
<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>
</ul>
</body></html>

```

Q6

Create 1 Public Docker Hub registry named cloudethix_hpa_yourname.

- Clone below repository on your system.

<https://github.com/vivekamin/kubernetes-hpa-example.git>

- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , build a docker image from the docker file and add meaningful tags and push to the docker hub repository.
- Once the image is pushed, go to k8s directory and update deployment.yaml file with image name from your repo. And then create it.
- Open service.yml and change the type to nodePort and apply the same.
- Open the HPA.yaml file, notice it and then apply the same.
- Open the browser, and access the webpage.
- Now it's time to test the HPA working with the below command.

```
kubectrl run -i --tty load-generator --rm --image=busybox --restart=Never --
/bin/sh -c "while sleep 0.01; do wget -q -O- http://NODE_PORT_SERVICE_NAME;
done"
```

- Check the HPA from kubectrl command and also check if the deployment is scaling up

```
avej_lanjekar@AvejLanjekar:Q6$ git clone
https://github.com/vivekamin/kubernetes-hpa-example.git
Cloning into 'kubernetes-hpa-example'...
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
avej_lanjekar@AvejLanjekar:Q6$ ls
kubernetes-hpa-example
avej_lanjekar@AvejLanjekar:Q6$ ls
kubernetes-hpa-example

avej_lanjekar@AvejLanjekar:Q6$ mkdir kubernetes-hpa
avej_lanjekar@AvejLanjekar:Q6$ cd kubernetes-hpa

avej_lanjekar@AvejLanjekar:Q6$ cp -r kubernetes-hpa-example/* kubernetes-hpa/

avej_lanjekar@AvejLanjekar:Q6$ ls
kubernetes-hpa  kubernetes-hpa-example

avej_lanjekar@AvejLanjekar:Q6$ cd kubernetes-hpa
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ ls
Dockerfile  README.md  k8s  package.json  src

avej_lanjekar@AvejLanjekar:kubernetes-hpa$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
```



```
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/avej_lanjekar/kubernetes-
Assignment/Repetition-1/Assignment3/Q6/kubernetes-hpa/.git/
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ git status
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
  (use "git add <file>..." to include in what will be committed)
```

```
    Dockerfile
    README.md
    k8s/
    package.json
    src/
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ ls
Dockerfile  README.md  k8s  package.json  src
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ docker build -t cloudehix_hpa .
STEP 1/6: FROM node:8.12.0-alpine
STEP 2/6: RUN mkdir -p /usr/src/app
--> Using cache
b408ab654063a834eb9496737462c2a8d2bd05d2a26be92b5d359bf125d5f883
--> b408ab654063
STEP 3/6: WORKDIR /usr/src/app
--> Using cache
7a294cff122ed4a38eefbae34788c1e2694b22804c54f3d9fa31494029d90f81
--> 7a294cff122e
STEP 4/6: COPY . /usr/src/app
--> bc221fe7d7e0
STEP 5/6: RUN npm install
```

```
npm notice created a lockfile as package-lock.json. You should commit this file.
```

```
npm WARN node-app@1.0.0 No repository field.
```

```
up to date in 10.705s
```

```
--> 6f6824c3001b
```

```
STEP 6/6: CMD ["npm", "start"]
```

```
COMMIT cloudehix_hpa
```

```
--> 787922fbbb15
```

```
Successfully tagged localhost/cloudehix_hpa:latest
```

```
787922fbbb153ea79d937e6dcd956e3b86e9ce8ccea7df1fe90f9674b14e08f7
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ docker tag cloudehix_hpa  
avejlanjekar45/cloudehix_hpa_avej
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ docker push  
avejlanjekar45/cloudehix_hpa_avej
```

```
Getting image source signatures
```

```
Copying blob 408b632b287b done |
```

```
Copying blob 43094e0eaa33 done |
```

```
Copying blob 4fe2ade4980c skipped: already exists
```

```
Copying blob e35f88fcc259 skipped: already exists
```

```
Copying blob eeb7d76f44e7 skipped: already exists
```

```
Copying blob b322a029158e skipped: already exists
```

```
Copying config 787922fbbb done |
```

```
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ ls
```

```
Dockerfile README.md k8s package.json src
```

```
avej_lanjekar@AvejLanjekar:kubernetes-hpa$ cd k8s/
```

```
avej_lanjekar@AvejLanjekar:k8s$ ls
```

```
deployment.yml hpa.yml service.yml
```

```
avej_lanjekar@AvejLanjekar:k8s$ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: node-example
```

```
spec:
  replicas: 1
  template:
    metadata:
      labels:
        app: node-example
    spec:
      containers:
      - name: node-example
        image: avejlanjekar45/cloudethix_hpa_avej
        imagePullPolicy: Always
        ports:
        - containerPort: 3000
      resources:
        limits:
          cpu: "0.5"
        requests:
          cpu: "0.25"
```

avej_lanjekar@AvejLanjekar:k8s\$ cat service.yml

```
apiVersion: v1
kind: Service
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  selector:
    app: node-example
  ports:
  - port: 3000
    protocol: TCP
    nodePort: 30001
  type: NodePort
```

avej_lanjekar@AvejLanjekar:k8s\$ kubectl apply -f service.yml
service/node-example created

avej_lanjekar@AvejLanjekar:k8s\$ cat hpa.yml

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  annotations:
  name: node-example
  namespace: default
spec:
```

```

maxReplicas: 4
minReplicas: 1
scaleTargetRef:
  apiVersion: extensions/v1
  kind: Deployment
  name: node-example
targetCPUUtilizationPercentage: 1

```

```

avej_lanjekar@AvejLanjekar:k8s$ kubectl apply -f hpa.yml
horizontalpodautoscaler.autoscaling/node-example created

```

```

avej_lanjekar@AvejLanjekar:k8s$ curl http://192.168.49.2:30001
Hello World!!

```

```

kubectl run -i --tty load-generator --rm --image=busybox --restart=Never --
/bin/sh -c "while sleep 0.01; do wget -q -O- http://node-example;
done"
/ # All commands and output from this session will be recorded in container
logs, including credentials and sensitive information passed through the
command prompt.
If you don't see a command prompt, try pressing enter.
/ #

```

```

avej_lanjekar@AvejLanjekar:~$ kubectl get hpa -w
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS
REPLICAS    AGE
node-example   Deployment/node-example   cpu: 0%/1%      1         4         1
4m40s
node-example   Deployment/node-example   cpu: 0%/1%      1         4         1
5m5s

```

Q7 →

- Create 1 Public Docker Hub registry named cloudeithx_cronjob_yourname.
- Initialize a local repository & copy below code (three files) to your local repository in any of your working branch.
- Once code is copied, build the docker image from Dockerfile , add meaningful tags and then push the docker image to Docker hub registry.
- Now update the pythoncronjob.yml file to change the image name that you have just pushed to docker hub registry.
- Now create a cron job using pythoncronjob.yml file. Check with kubectl command if the cron job is created.

- Check the Job name which is created by cronjob from command line or lens.
- Then check the pod logs which are created by the job and capture the output

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ ls
Dockerfile  helloworld.py  pythoncronjob.yml
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ cat Dockerfile
FROM python:3.7-alpine
#add user group and ass user to that group
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
#creates work dir
WORKDIR /app
#copy python script to the container folder app
COPY helloworld.py /app/helloworld.py
RUN chmod +x /app/helloworld.py
#user is appuser
USER appuser
ENTRYPOINT ["python", "/app/helloworld.py"]
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ cat helloworld.py
#!/usr/local/bin/python3
import datetime
x = datetime.datetime.now()
print("Welcome to the Cloudethix World")
print("Today is")
print(x)
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ cat pythoncronjob.yml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: python-helloworld
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: python-helloworld
              image: avejlanjekar45/cloudethix_cronjob_avej
              command: ["python", "/app/helloworld.py"]
          restartPolicy: OnFailure
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ docker build -t cronjob .
```

```
STEP 1/7: FROM python:3.7-alpine
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:3.7-alpine...
Getting image source signatures
Copying blob 96526aa774ef done    |
Copying blob ea1518237b37 done    |
Copying blob 9875af95546d done    |
Copying blob 4819c95424fc done    |
Copying blob 148762f75a1f done    |
Copying config 1bac8ae77e done    |
Writing manifest to image destination
STEP 2/7: RUN addgroup -S appgroup && adduser -S appuser -G appgroup
--> 38b117328542
STEP 3/7: WORKDIR /app
--> 6438ab034fa9
STEP 4/7: COPY helloworld.py /app/helloworld.py
--> 63e924043b9b
STEP 5/7: RUN chmod +x /app/helloworld.py
--> a92c7ffe4a8c
STEP 6/7: USER appuser
--> 07de10ce0353
STEP 7/7: ENTRYPOINT ["python", "/app/helloworld.py"]
COMMIT cronjob
--> c41e700fdbd7
Successfully tagged localhost/cronjob:latest
c41e700fdbd76e0d5f291dbc4565caeec8cc244dfbddd17cf411ef37ac23c0ac
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ docker tag cronjob
avejlanjekar45/cloudethix_cronjob_avej
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ docker push
avejlanjekar45/cloudethix_cronjob_avej
Getting image source signatures
Copying blob 5795f33d6724 done    |
Copying blob 148762f75a1f skipped: already exists
Copying blob ea1518237b37 skipped: already exists
Copying blob 4819c95424fc skipped: already exists
Copying blob 734bd37c36be done    |
Copying blob 96526aa774ef skipped: already exists
Copying blob 734bd37c36be done    |
Copying blob 9875af95546d skipped: already exists
Copying config c41e700fdb done    |
Writing manifest to image destination
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ kubectl apply -f
pythoncronjob.yml
cronjob.batch/python-helloworld created
```

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ kubectl get jobs -w
```

NAME	STATUS	COMPLETIONS	DURATION	AGE	
python-helloworld-29807008	Running	0/1		0s	
python-helloworld-29807008	Running	0/1	0s	0s	
python-helloworld-29807008	SuccessCriteriaMet	0/1		18s	18s
python-helloworld-29807008	Complete	1/1		17s	18s
python-helloworld-29807009	Running	0/1			0s
python-helloworld-29807009	Running	0/1		0s	0s
python-helloworld-29807009	SuccessCriteriaMet	0/1		5s	5s
python-helloworld-29807009	Complete	1/1		5s	5s
python-helloworld-29807010	Running	0/1			0s
python-helloworld-29807010	Running	0/1		0s	0s
python-helloworld-29807010	SuccessCriteriaMet	0/1		6s	6s
python-helloworld-29807010	Complete	1/1		6s	6s

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS
python-helloworld-29807008-fd8kr	0/1	Completed	0
2m46s			
python-helloworld-29807009-frxgq	0/1	Completed	0
106s			
python-helloworld-29807010-kxb9m	0/1	Completed	0
46s			

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$ kubectl get jobs
```

NAME	STATUS	COMPLETIONS	DURATION	AGE
python-helloworld-29807009	Complete	1/1	5s	2m14s
python-helloworld-29807010	Complete	1/1	6s	74s
python-helloworld-29807011	Complete	1/1	6s	14s

```
avej_lanjekar@AvejLanjekar:cloudethix_cronjob$
```